João André **Roque Costa**

Software, Data & AI Engineer

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Software engineer with an MSc in Computer Science and Engineering, focused on **Python**, SQL, statistical analysis, machine learning, and dashboards. Built event-driven services, data workflows, scalable algorithms, and operational dashboards. Interested in applying software engineering and analytics to telemetry, motion, and simulation systems.

Professional Experience

Deloitte — *Coimbra, Portugal*

Tech Analyst — *Apr. 2026 - Oct. 2026*

- **Built** a cross-functional **Python**/LLM prototype for data migration and synthetic-data generation, integrating Azure AI Foundry, Blob Storage, MongoDB, and the Claude SDK.
- **Automated** data-oriented reporting with Python and Playwright, and contributed RAG/OCR workflow designs for document analysis and structured data extraction.
- **Delivered** an Azure subscription migration for Databricks and Data Factory assets, coordinating dependent services across a cloud data platform.

Instituto Superior Técnico — *Lisbon, Portugal*

Teaching Assistant — *Oct. 2024 - Jul. 2025*

- **Taught** and evaluated practical classes in Databases and Data Science, explaining technical concepts and adapting communication to different audiences.
- **Guided** Python work in data preparation, statistical analysis, and visualization with pandas and matplotlib, emphasizing reproducible analysis and clear communication of results.

Education

Instituto Superior Técnico — *Lisbon, Portugal*

M.Sc. Computer Science and Engineering — *Sep 2023 - Nov 2025*

- Focused on **Artificial Intelligence** and **Data Science**, applying statistical and machine-learning foundations to data-driven modelling.
- **Relevant Coursework**: Deep Learning, Data Science, Multi-Agent Systems, Optimization Algorithms, Cloud Computing.
- Thesis: **Maximum Variance Unfolding on Disjoint Manifolds (MVU-DM)**.
  - Developed a decomposition-based extension to MVU for **nonlinear dimensionality reduction** on disconnected datasets.
  - Designed a parallelizable method for large datasets, preserving local geometry while producing comparable embeddings.

Instituto Superior Técnico — *Lisbon, Portugal*

B.Sc. Computer Science and Engineering — *Sep 2020 - Jul 2023*

- Studied algorithms, distributed systems, operating systems, and database theory, building foundations for maintainable software and data-intensive systems.
- Applied **C/C++** in systems projects and designed relational schemas and transactions in **SQL**.

Extracurricular

Personal Software Projects (Self-Hosted)

- Built a self-hosted Python monitoring service with historical trend analysis, alerts, and an operational dashboard for tracking system state.
- Developed an event-driven Discord matchmaking service using ELO ratings, PostgreSQL match history, and Redis queues and caching.
- Built automation and API integrations for multi-platform price monitoring, risk management, optimal stake sizing, and inventory tracking.

Skills & Competencies

Software Engineering	**Python** (pandas, NumPy, scikit-learn, TensorFlow, matplotlib, Playwright), SQL, C/C++, REST APIs
Data, AI & Statistics	Data analysis, statistical modelling, machine learning, deep learning, LLMs, RAG, trend analysis
Dashboards & Automation	Python automation, web interfaces, operational dashboards, monitoring dashboards
Cloud & Data Platforms	Azure AI Foundry, Azure, Databricks, Data Factory, Blob Storage, Managed Instances, Claude SDK
Data Systems	PostgreSQL, MongoDB, MySQL, Redis, SQL Server, data pipelines, historical analysis, caching and queues
Languages	Portuguese (Native), English (Fluent)
Collaboration	Cross-functional communication, mentoring, technical presentation, communicating analysis